॥वसुधैव कुटुम्बकम्॥

SYMBIOSIS INSTITUTE OF TECHNOLOGY

(Established under section 3 of the UGC Act 1956) Re-accredited by NAAC with “A” Grade

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Batch – B3 COURSE – JAVA

Experiment 9-

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

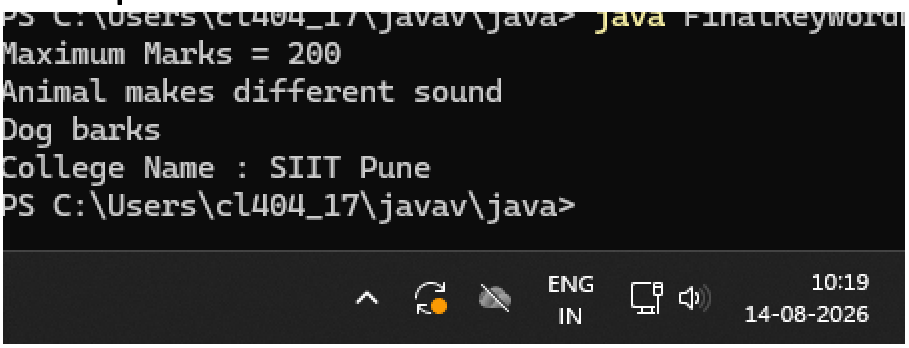
AIM – Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Program 9

```java
// Program to demonstrate the final keyword in Java

// Final class (cannot be inherited)
final class College {

    void displayCollege() {
        System.out.println("College Name : SIIT Pune");
    }
}

// Parent class
class Animal {

    // Final variable (constant)
    final int MAX_MARKS = 200;

    // Final method (cannot be overridden)
    final void sound() {
        System.out.println("Animal makes different sound");
    }

    void displayMarks() {
        System.out.println("Maximum Marks = " + MAX_MARKS);
    }
}

// Child class
class Dog extends Animal {

    void displayDog() {
        System.out.println("Dog barks");
    }

    /*
    // This is NOT allowed because sound() is final.
    // Uncommenting it will give a compile-time error.

    void sound() {
        System.out.println("Dog barks");
    }
    */
}

// Main class
public class FinalKeywordDemo {
```

```java
    public static void main(String[] args) {

        Dog d = new Dog();

        d.displayMarks();
        d.sound();
        d.displayDog();

        College c = new College();
        c.displayCollege();
    }
}
```

Output



```
PS C:\Users\cl404_17\javav\java> java FinalKeyword
Maximum Marks = 200
Animal makes different sound
Dog barks
College Name : SIIT Pune
PS C:\Users\cl404_17\javav\java>
```


EXERCISE

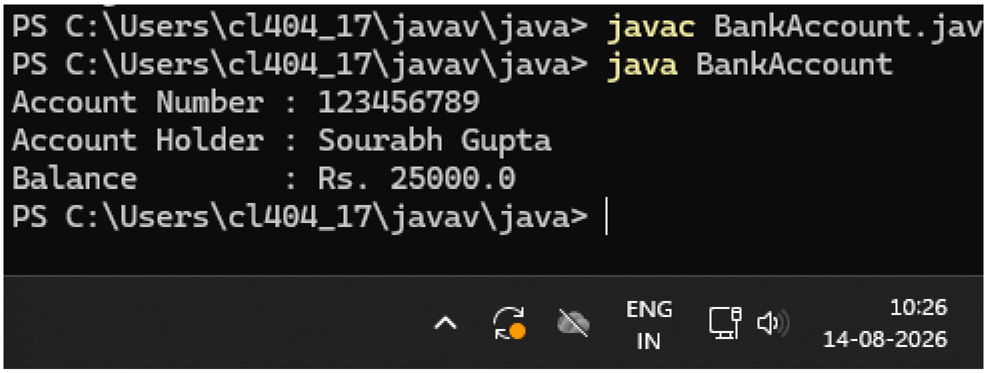
9.1 . Create a Bank Account program where account number is final and cannot be changed

Program 9.1 -

```
class BankAccount {

    // Final variable
    final int accountNumber;

    String accountHolder;
    double balance;

    // Constructor
    BankAccount(int accountNumber, String accountHolder, double balance) {
        this.accountNumber = accountNumber;
        this.accountHolder = accountHolder;
        this.balance = balance;
    }

    // Method to display account details
    void display() {
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Account Holder : " + accountHolder);
        System.out.println("Balance        : Rs. " + balance);
    }

    public static void main(String[] args) {

        BankAccount account = new BankAccount(
            123456789,
            "Sourabh Gupta",
            25000
        );

        account.display();

        // account.accountNumber = 987654321;
        // Error: cannot assign a value to final variable accountNumber
```

```
    }
}
```

Screenshot 9.1 –



Program 9.2- Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```java
class LibraryBook {

    // Final variable
    final String ISBN;

    String title;
    String author;
    double price;

    // Constructor
```

```java
    LibraryBook(String ISBN, String title, String author, double price) {
        this.ISBN = ISBN;
        this.title = title;
        this.author = author;
        this.price = price;
    }

    // Method to display book details
    void display() {
        System.out.println("ISBN   : " + ISBN);
        System.out.println("Title  : " + title);
        System.out.println("Author : " + author);
        System.out.println("Price  : Rs. " + price);
    }

    public static void main(String[] args) {

        LibraryBook book = new LibraryBook(
            "978-81-12345-67-8",
            "Java Programming",
            "James Gosling",
            599
        );

        book.display();

        // ISBN cannot be changed because it is final
        // book.ISBN = "978-81-98765-43-2";
    }
}
```

Screenshot 9.2 –

```
PS C:\Users\cl404_17\javav\java> javac LibraryBook.j
PS C:\Users\cl404_17\javav\java> java LibraryBook
ISBN   : 978-81-12345-67-8
Title  : Java Programming
Author : James Gosling
Price  : Rs. 599.0
PS C:\Users\cl404_17\javav\java>
```